

COLON NUCLEI SEGMENTATION USING IMAGE ANALYSIS AND DEEP LEARNING

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by

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Declaration

I hereby declare that the work presented in this report entitled "**Colon Nuclei Segmentation Using Image Analysis and Deep Learning**" is an original work carried out by me under the supervision of **Dr. Himanshu Pramod Padole**, School of Electrical and Computer Sciences, Indian Institute of Technology Bhubaneswar.

This work has not been submitted elsewhere for the award of any degree, diploma, or any other academic qualification.

Yedupati Harsha Vardhan
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Supervisor Signature

Acknowledgement

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Abstract

Accurate nuclei segmentation in colon histopathological images plays a crucial role in quantitative tissue analysis and computer-aided diagnosis. It enables the study of nuclear morphology, cellular density, and tissue architecture, which are essential for pathological assessment. However, automated nuclei segmentation remains challenging due to overlapping nuclei, irregular boundaries, staining variations, and densely packed cellular structures.

This work presents a comparative study of traditional image processing techniques and deep learning-based approaches for nuclei segmentation using the CoNIC dataset. Initially, a watershed-based segmentation pipeline was implemented as a classical baseline approach involving thresholding, morphological operations, distance transformation, and marker-controlled watershed segmentation.

To improve segmentation performance, deep learning architectures including U-Net, U-Net++, and Scale Aggregation Network (SANet) were implemented and evaluated. Further improvements were introduced by incorporating attention mechanisms into U-Net++ to improve feature selection during skip connections. A hybrid loss function combining Binary Cross Entropy loss, Dice loss, and Focal loss was used to improve segmentation robustness and address class imbalance.

To further enhance feature extraction, a multiscale attention mechanism using multiple convolution kernels and a JPFM-inspired dilated bottleneck module were integrated to capture both local and global contextual information. Experimental evaluation demonstrated that the final architecture achieved the best performance with a Dice score of 0.8074 and an Intersection over Union (IoU) score of 0.7177.

The results demonstrate that progressive architectural improvements can significantly enhance nuclei segmentation performance in colon histopathological images.

Keywords: Colon Histopathology, Nuclei Segmentation, Deep Learning, U-Net, Attention Mechanism, Multiscale Feature Extraction

Contents

1	Introduction	1
1.1	Background	1
1.2	Literature Survey	1
1.3	Motivation	2
2	Dataset and Preprocessing	3
2.1	CoNIC Dataset	3
2.2	Data Preparation	4
2.3	Train-Test Split	4
2.4	Evaluation Metrics	4
2.4.1	Intersection over Union (IoU)	4
2.4.2	Dice Score	5
2.4.3	Precision	5
2.4.4	Recall	5
2.4.5	Count MAE	5
3	Methodology	6
3.1	Overview of Methodology	6
3.2	Watershed Segmentation	6
3.3	U-Net Architecture	7
3.4	U-Net++ Architecture	8
3.5	Scale Aggregation Network (SANet)	9
3.6	Attention U-Net++	10
3.6.1	Dice Loss	11
3.6.2	Focal Loss	11
3.6.3	Hybrid Loss	11
3.7	Multiscale Attention + JPFM Enhancement	11

3.7.1	Multiscale Attention	11
3.7.2	JPFM Bottleneck	12
3.8	Training Configuration	12
4	Experimental Results and Discussion	14
4.1	Experimental Setup	14
4.2	Evaluation Metrics	15
4.2.1	Intersection over Union	15
4.2.2	Dice Score	15
4.2.3	Precision	15
4.2.4	Recall	15
4.2.5	Count Mean Absolute Error	15
4.3	Qualitative Analysis	16
4.4	Training Performance Analysis	17
4.5	Quantitative Performance Analysis	17
4.6	Final Performance Comparison	18
4.6.1	Extended Training Analysis of Proposed Model	19
4.7	Discussion	22
4.8	Summary	22
5	Conclusion and Future Scope	23
5.1	Conclusion	23
5.2	Limitations	24
5.3	Future Scope	24
6	Bibliography	26
6.1	Dataset Reference	26
6.2	Research Papers	26
6.3	Software Libraries	27

List of Figures

3.1	Watershed segmentation pipeline	7
3.2	U-Net architecture	8
3.3	U-Net++ architecture	9
3.4	SANet architecture	10
3.5	Attention U-Net++ architecture	11
3.6	Multiscale Attention + JPFM enhanced U-Net++	12
4.1	Qualitative comparison of all implemented models with nuclei count estimation	16
4.2	Training performance comparison of all deep learning models	17
4.3	Quantitative performance comparison of all models	18
4.4	Segmentation and nuclei counting results after extended training . .	21

List of Tables

4.1	Training Configuration	15
4.2	Final Performance Comparison	18
4.3	Performance of Proposed Model after Extended Training	20
4.4	Comparison of Original and Extended Training Performance	20

Chapter 1

Introduction

1.1 Background

Histopathological image analysis plays a significant role in modern medical diagnosis by enabling detailed examination of tissue structures at the cellular level. Among various tasks in histopathology, nuclei segmentation is one of the most fundamental steps for quantitative tissue analysis. Accurate segmentation of nuclei helps pathologists analyze nuclear morphology, tissue organization, cellular density, and abnormal growth patterns.

Colon histopathological images present unique challenges for automated nuclei segmentation due to overlapping nuclei, irregular nuclear boundaries, staining variations, and densely packed cellular regions. Manual annotation of nuclei is time-consuming and prone to inter-observer variability, making automated segmentation techniques highly desirable.

Traditional image processing techniques and deep learning approaches have both been explored to solve this problem.

1.2 Literature Survey

Early nuclei segmentation methods relied on traditional image processing techniques such as thresholding, edge detection, morphological operations, and watershed segmentation. While these approaches were computationally simple, they often failed in cases involving overlapping nuclei, noisy tissue regions, and irregular boundaries.

The introduction of U-Net significantly improved biomedical image segmentation

by using an encoder-decoder architecture with skip connections for preserving spatial information.

U-Net++ further enhanced segmentation performance through nested skip pathways that reduced the semantic gap between encoder and decoder feature maps.

Attention-based segmentation architectures improved localization by allowing the model to focus on important nuclei regions while suppressing irrelevant background features.

Scale Aggregation Network (SANet) introduced multi-scale convolutions for density estimation and nuclei counting tasks.

Motivated by the strengths and limitations of these models, this project implements multiple architectures and proposes a MultiScale Attention + Joint Pyramid Fusion Module (JPFM) model for improved colon nuclei segmentation.

1.3 Motivation

Despite significant progress in nuclei segmentation, several challenges still remain in colon histopathological image analysis:

- Overlapping nuclei
- Irregular boundaries
- Dense cellular clusters
- Variations in staining intensity
- Difficulty in capturing both local and global contextual information

Traditional approaches often suffer from over-segmentation and poor robustness. Although deep learning models provide improved performance, standard architectures may still struggle to accurately segment complex nuclear structures.

This motivated the exploration of progressively improved architectures starting from watershed segmentation, followed by U-Net, U-Net++, attention-based improvements, and finally multiscale feature aggregation techniques for achieving better segmentation performance.

Chapter 2

Dataset and Preprocessing

2.1 CoNIC Dataset

The Colon Nuclei Identification and Counting (CoNIC) dataset was used in this work for nuclei segmentation experiments. The dataset consists of histopathological image patches extracted from colon tissue samples with manually annotated nuclei masks.

The dataset contains images with significant variations in tissue structures, nuclear shapes, staining patterns, and cellular density. These variations make the dataset highly suitable for evaluating segmentation algorithms under realistic conditions.

Each image patch contains corresponding ground truth annotations that provide pixel-level nuclear segmentation masks.

The dataset presents several challenges such as:

- Overlapping nuclei
- Dense nuclear clusters
- Irregular nuclear boundaries
- Variations in staining
- Presence of background artifacts

These characteristics make CoNIC a challenging benchmark for automated nuclei segmentation.

2.2 Data Preparation

The dataset was prepared before training the segmentation models.

- Input histopathology images were resized to 256×256 pixels.
- Corresponding segmentation masks were extracted from annotation files.
- Binary masks were generated for segmentation experiments.
- Images were normalized for stable deep learning training.

The final processed dataset consisted of image-mask pairs suitable for both traditional and deep learning experiments.

2.3 Train-Test Split

The dataset was divided into training and testing sets using an 80:20 ratio.

Additionally, 10% of the training data was used as validation data during training.

- Training set: 3984 samples used for model learning
- Testing set: 997 samples used for final evaluation
- Validation split: 10% used for monitoring model performance

2.4 Evaluation Metrics

The performance of all models was evaluated using the following metrics:

2.4.1 Intersection over Union (IoU)

IoU measures the overlap between predicted segmentation masks and ground truth masks.

$$IoU = \frac{TP}{TP + FP + FN} \quad (2.1)$$

2.4.2 Dice Score

Dice score measures segmentation similarity between predicted and ground truth masks.

$$Dice = \frac{2TP}{2TP + FP + FN} \quad (2.2)$$

2.4.3 Precision

Precision measures the proportion of correctly predicted nuclei pixels among all predicted positive pixels.

$$Precision = \frac{TP}{TP + FP} \quad (2.3)$$

2.4.4 Recall

Recall measures the proportion of correctly detected nuclei pixels.

$$Recall = \frac{TP}{TP + FN} \quad (2.4)$$

2.4.5 Count MAE

Mean Absolute Error (MAE) was used for nuclei counting evaluation.

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (2.5)$$

Chapter 3

Methodology

3.1 Overview of Methodology

The objective of this work was to progressively improve nuclei segmentation performance in colon histopathological images by evaluating multiple traditional and deep learning-based approaches. The methodology began with conventional image processing techniques and gradually moved toward advanced deep learning architectures with architectural improvements.

The workflow followed in this project is shown below:

- Watershed Segmentation
- U-Net
- U-Net++
- SANet
- Attention U-Net++
- Multiscale Attention + JPFM Enhanced U-Net++

Each model was trained and evaluated independently using the same dataset for fair comparison.

3.2 Watershed Segmentation

Watershed segmentation was implemented as the traditional baseline model. This approach treats grayscale images as topographical surfaces where pixel intensities

represent elevation values.

The following steps were performed:

1. RGB to grayscale conversion
2. Otsu thresholding
3. Morphological opening
4. Distance transform
5. Marker extraction
6. Watershed segmentation

Distance transformation was used to identify nucleus centers:

$$D(x, y) = \min_{(i, j) \in B} \sqrt{(x - i)^2 + (y - j)^2} \quad (3.1)$$

where B represents background pixels.

Although computationally efficient, watershed segmentation struggled with overlapping nuclei and produced over-segmentation in dense regions.

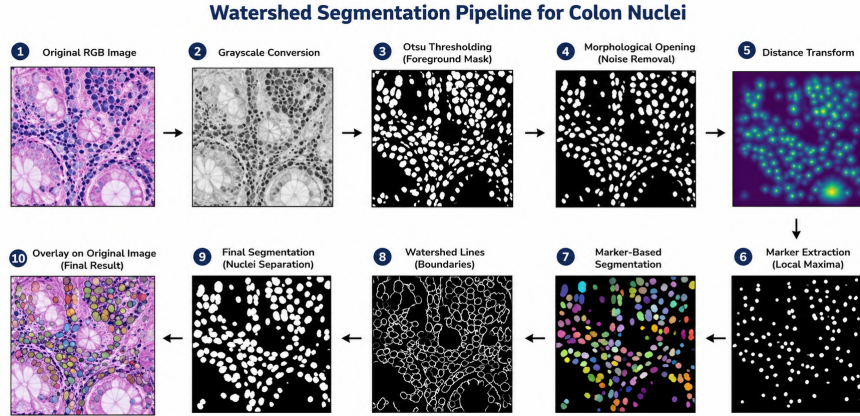


Figure 3.1: Watershed segmentation pipeline

3.3 U-Net Architecture

U-Net was implemented as the first deep learning segmentation baseline.

The architecture consists of:

- Encoder path
- Bottleneck layer
- Decoder path
- Skip connections

The encoder extracts hierarchical features through repeated convolution and pooling operations. The decoder reconstructs segmentation masks through upsampling operations.

Skip connections preserve spatial information lost during downsampling.

The model was trained using Binary Cross Entropy loss:

$$L_{BCE} = -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})] \quad (3.2)$$

where:

- y = ground truth mask
- $\hat{y}$ = predicted mask

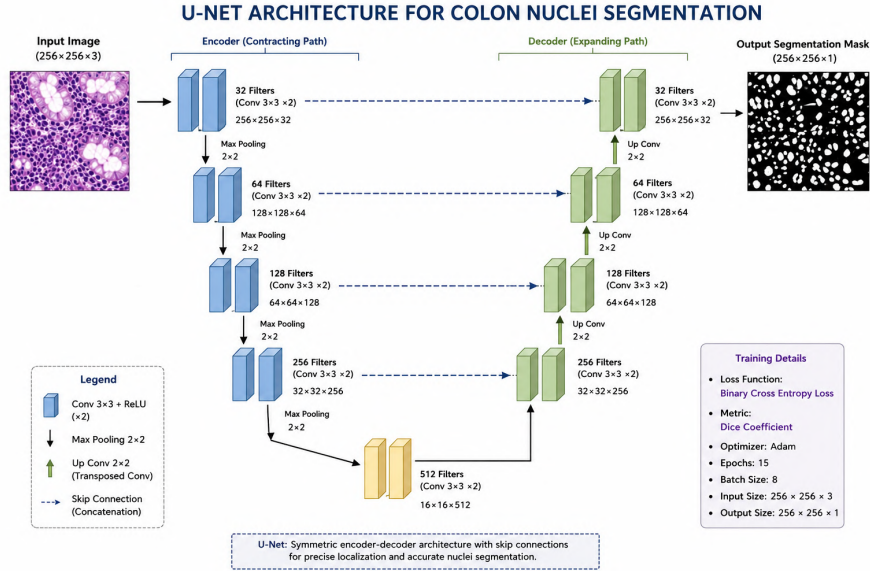


Figure 3.2: U-Net architecture

3.4 U-Net++ Architecture

U-Net++ was implemented to improve feature fusion between encoder and decoder layers.

In standard U-Net, there exists a semantic gap between encoder and decoder feature maps. U-Net++ reduces this gap by introducing nested skip pathways and dense convolution blocks.

These nested connections improve feature propagation and enable better reconstruction of nuclei boundaries.

The model was also trained using Binary Cross Entropy loss:

$$L_{BCE} = -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})] \quad (3.3)$$

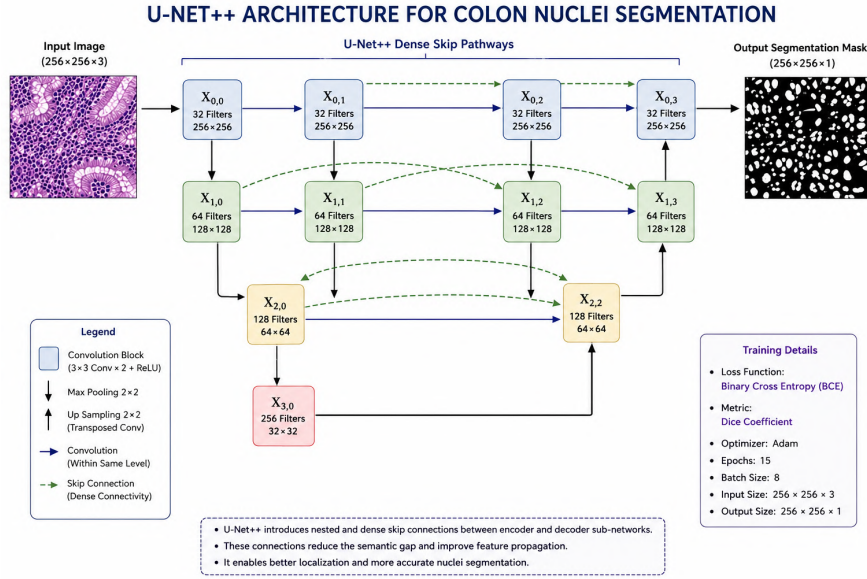


Figure 3.3: U-Net++ architecture

3.5 Scale Aggregation Network (SANet)

SANet was implemented for nuclei counting using density estimation.

Instead of generating segmentation masks directly, SANet predicts density maps where integration of pixel values provides nuclei count estimation.

The density map generation process can be represented as:

$$D(x) = \sum_{i=1}^N \delta(x - x_i) * G_{\sigma}(x) \quad (3.4)$$

where:

- x_i = nucleus centroid

- G_σ = Gaussian kernel

The model was trained using Mean Squared Error loss:

$$L_{MSE} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (3.5)$$

Although SANet performed well in counting tasks, it was less suitable for precise boundary segmentation.

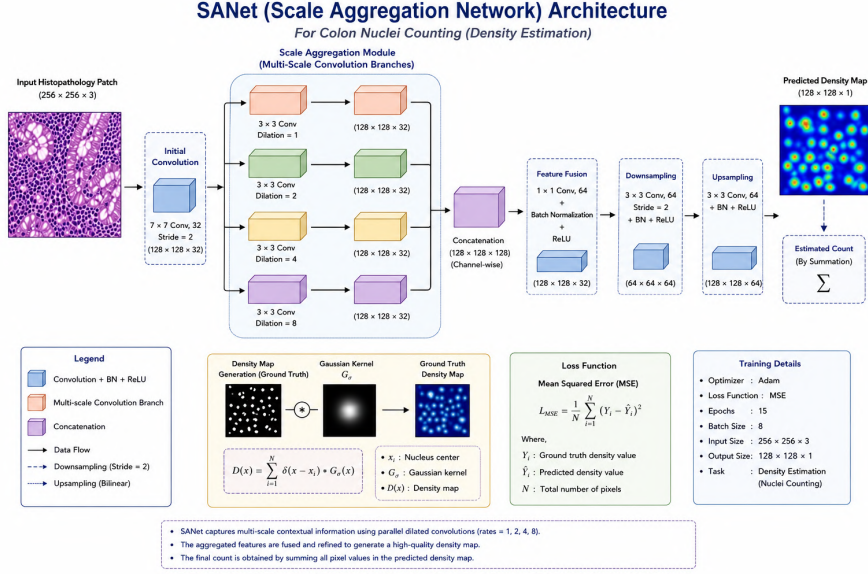


Figure 3.4: SANet architecture

3.6 Attention U-Net++

To improve segmentation performance further, attention gates were integrated into U-Net++.

These gates help the model focus on relevant nuclei regions while suppressing background noise.

Attention gates were integrated into U-Net++ skip connections to improve feature selection.

These gates suppress irrelevant background features and improve nuclei localization.

The model was trained using a hybrid loss consisting of Binary Cross Entropy, Dice Loss and Focal Loss.

3.6.1 Dice Loss

$$L_{Dice} = 1 - \frac{2|Y \cap \hat{Y}| + \epsilon}{|Y| + |\hat{Y}| + \epsilon} \quad (3.6)$$

3.6.2 Focal Loss

$$L_{Focal} = -\alpha(1 - p_t)^\gamma \log(p_t) \quad (3.7)$$

3.6.3 Hybrid Loss

$$L = 0.3L_{BCE} + 0.4L_{Dice} + 0.3L_{Focal} \quad (3.8)$$

This improved segmentation robustness and addressed class imbalance.

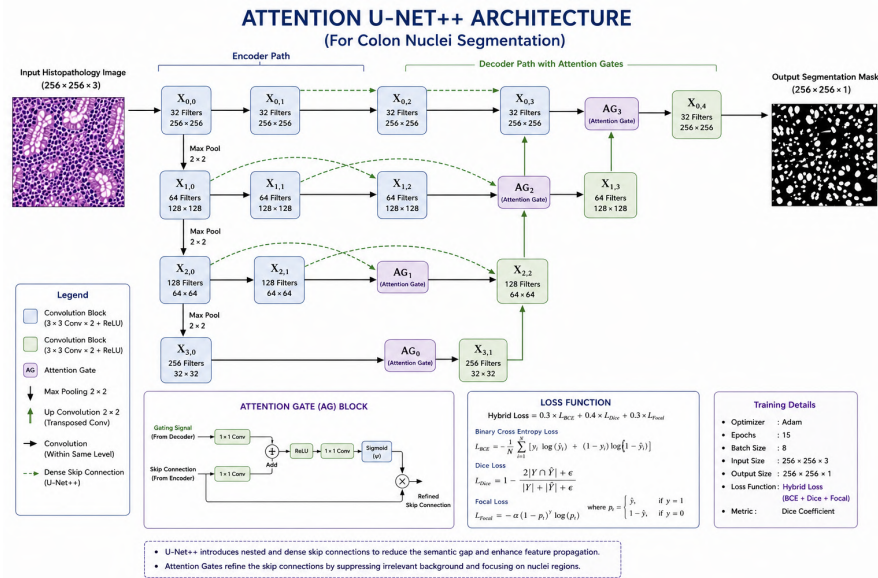


Figure 3.5: Attention U-Net++ architecture

3.7 Multiscale Attention + JPFM Enhancement

To further improve segmentation performance, additional architectural enhancements were introduced.

3.7.1 Multiscale Attention

Multiple convolution kernels were used:

$$1 \times 1, 3 \times 3, 5 \times 5 \quad (3.9)$$

These kernels capture both fine-grained and broader contextual features.

3.7.2 JPFM Bottleneck

Dilated convolutions were introduced to capture wider contextual information:

$$d = 1, 2, 4, 8 \quad (3.10)$$

The outputs from all branches were concatenated before decoding.

The final model was trained using the same hybrid loss:

$$L = 0.3L_{BCE} + 0.4L_{Dice} + 0.3L_{Focal} \quad (3.11)$$

This model achieved the best segmentation performance.

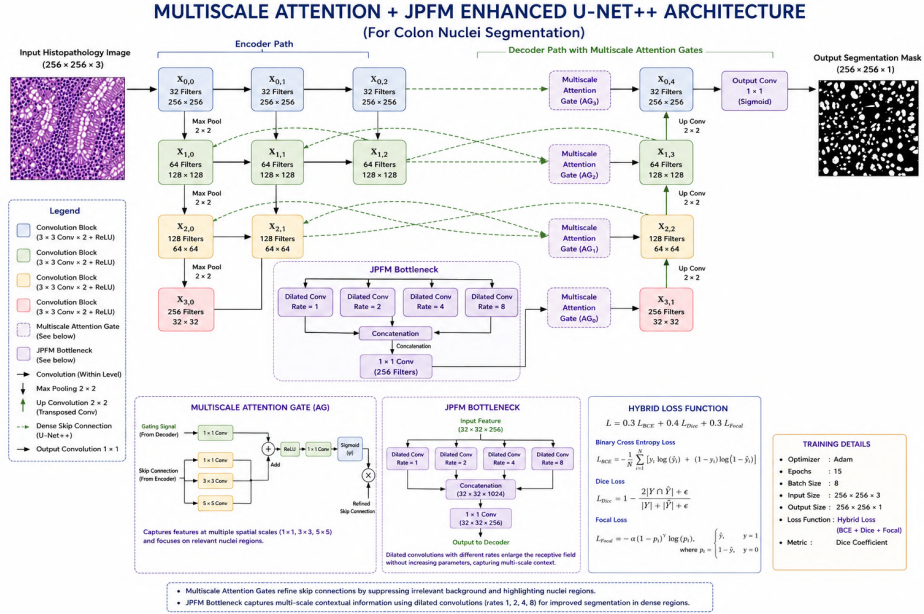


Figure 3.6: Multiscale Attention + JPFM enhanced U-Net++

3.8 Training Configuration

All deep learning models were trained using the following configuration:

- Input size: 256×256

- Epochs: 15
- Batch size: 8
- Optimizer: Adam
- GPU acceleration

The final evaluation was performed using IoU, Dice Score, Precision, Recall, and Count MAE.

Chapter 4

Experimental Results and Discussion

This chapter presents the experimental evaluation of all implemented approaches for colon nuclei segmentation and counting using the CoNIC dataset. Both traditional image processing methods and deep learning architectures were evaluated through qualitative and quantitative analysis.

The following approaches were implemented:

- Watershed Segmentation
- U-Net
- U-Net++
- SAnet
- Attention U-Net++
- MultiScale Attention + JPFM Enhanced Model

The objective of this chapter is to analyze the strengths and limitations of each approach and identify the best-performing model.

4.1 Experimental Setup

All experiments were conducted using Python with TensorFlow/Keras, OpenCV, NumPy, and Matplotlib libraries.

The dataset was divided into training and testing sets using an 80:20 ratio.

Training configuration is shown in Table 4.1.

Table 4.1: Training Configuration

Parameter	Value
Input Size	$256 \times 256 \times 3$
Train-Test Split	80:20
Batch Size	8
Epochs	15
Optimizer	Adam

4.2 Evaluation Metrics

The following evaluation metrics were used:

4.2.1 Intersection over Union

$$IoU = \frac{TP}{TP + FP + FN} \quad (4.1)$$

4.2.2 Dice Score

$$Dice = \frac{2TP}{2TP + FP + FN} \quad (4.2)$$

4.2.3 Precision

$$Precision = \frac{TP}{TP + FP} \quad (4.3)$$

4.2.4 Recall

$$Recall = \frac{TP}{TP + FN} \quad (4.4)$$

4.2.5 Count Mean Absolute Error

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (4.5)$$

4.3 Qualitative Analysis

A consolidated qualitative comparison was performed using representative test samples from the CoNIC dataset.

Figure 4.1 shows:

- Original image
- Ground truth segmentation mask
- Ground truth nuclei count
- Predicted outputs from all implemented models
- Predicted nuclei counts

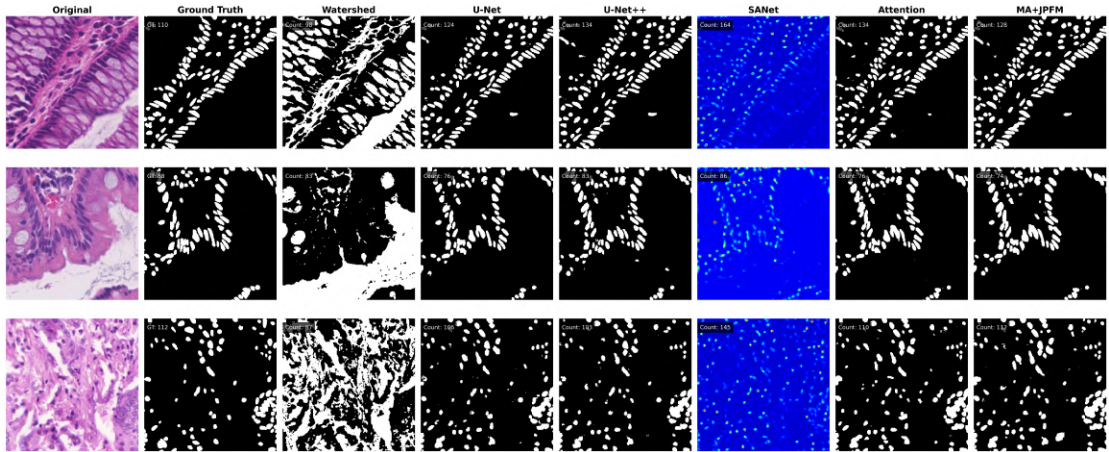


Figure 4.1: Qualitative comparison of all implemented models with nuclei count estimation

From the qualitative results:

- Watershed frequently over-segmented clustered nuclei.
- U-Net improved segmentation quality significantly.
- U-Net++ improved boundary refinement.
- SANet produced density maps for counting but lacked precise boundaries.
- Attention U-Net++ improved localization.
- MultiScale Attention+JPFM produced the most refined segmentation masks.

4.4 Training Performance Analysis

Training curves of all deep learning models were consolidated into a single subplot figure for better comparison.

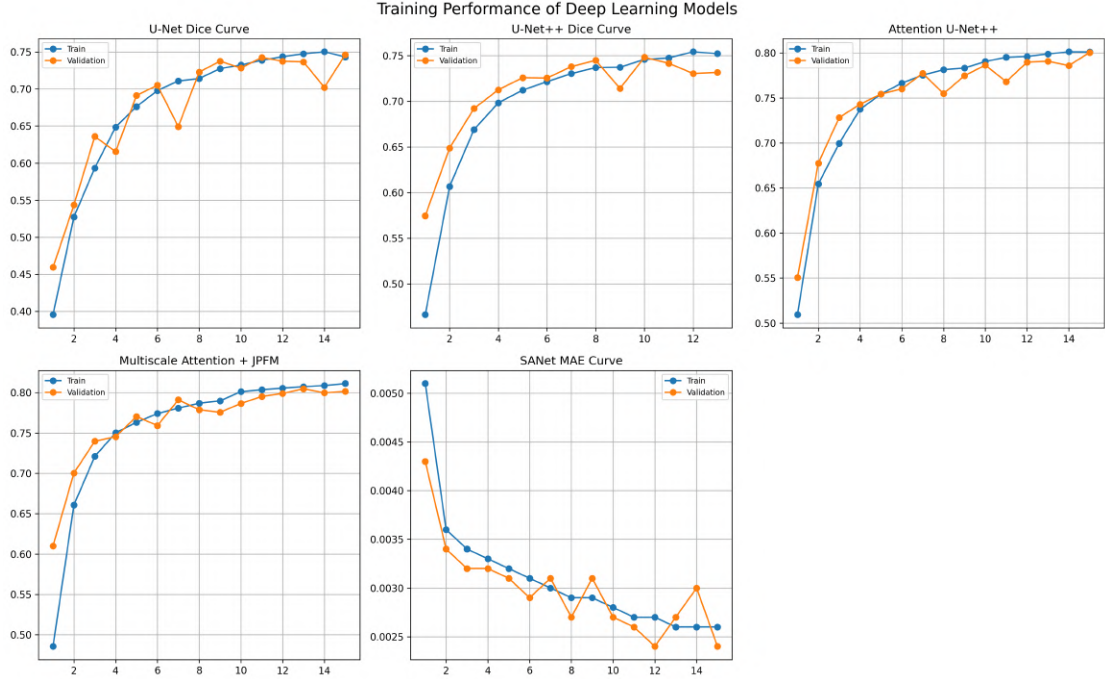


Figure 4.2: Training performance comparison of all deep learning models

The following observations were made:

- U-Net showed stable convergence.
- U-Net++ showed slightly improved learning stability.
- SANet minimized counting loss effectively.
- Attention U-Net++ achieved higher validation Dice scores.
- MultiScale Attention + JPFM showed the best convergence trend.

4.5 Quantitative Performance Analysis

The quantitative performance comparison is shown in Figure 4.3.

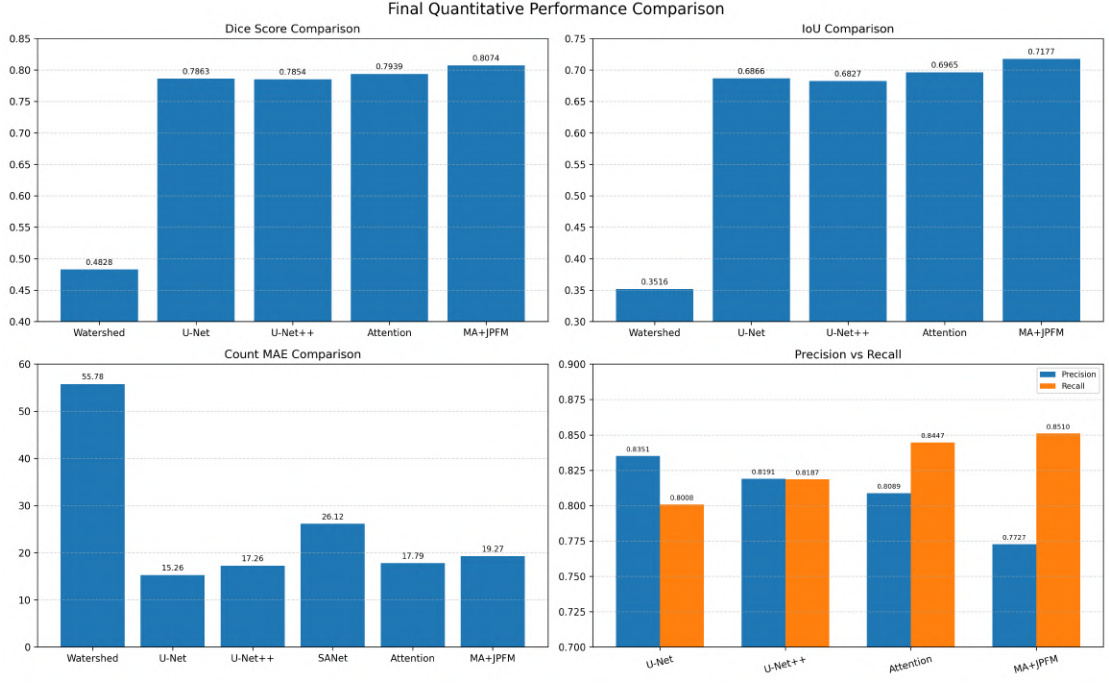


Figure 4.3: Quantitative performance comparison of all models

The subplot includes:

- Dice Score comparison
- IoU comparison
- Precision vs Recall comparison
- Count MAE comparison

4.6 Final Performance Comparison

Table 4.2 summarizes the final performance of all models.

Table 4.2: Final Performance Comparison

Model	IoU	Dice	Precision	Recall	Count MAE
Watershed	0.3516	0.4828	—	—	55.78
U-Net	0.6866	0.7863	0.8351	0.8008	15.26
U-Net++	0.6827	0.7854	0.8191	0.8187	17.26
SANet	—	—	—	—	26.12
Attention U-Net++	0.6965	0.7939	0.8089	0.8447	17.79
MultiScale + JPFM	0.7177	0.8074	0.7727	0.8510	19.27

The MultiScale Attention + JPFM model achieved the highest IoU, Dice Score, and Recall among all segmentation models. Although the model achieved the highest IoU, Dice score, and recall, its precision was slightly lower due to aggressive nuclei detection that introduced minor false positives.

Although U-Net achieved lower Count MAE, the final model demonstrated superior segmentation quality.

4.6.1 Extended Training Analysis of Proposed Model

To further analyze the performance of the proposed MultiScale Attention + JPFM model, an additional experiment was conducted by increasing the maximum training epochs from 15 to 100. The objective of this experiment was to determine whether extended training could further improve segmentation accuracy.

Unlike the original experiment, this training setup incorporated additional optimization strategies such as:

- Early Stopping
- ReduceLROnPlateau
- Model Checkpointing

These techniques were introduced to prevent overfitting and retain the best-performing model during training.

Initially, the model showed consistent improvement in validation Dice score during the early epochs. The highest validation Dice score was achieved at Epoch 19:

$$Dice = 0.8028$$

After Epoch 19, training Dice continued increasing while validation performance started fluctuating, indicating the beginning of overfitting. At Epoch 27, early stopping was triggered and the model restored weights from the best-performing epoch.

The final test performance obtained from the extended training experiment is shown in Table 4.3.

Table 4.3: Performance of Proposed Model after Extended Training

Metric	Value
Loss	0.1356
Accuracy	0.9421
Dice Score	0.8010

Table 4.4 compares the original proposed model with the extended training experiment.

Table 4.4: Comparison of Original and Extended Training Performance

Model Version	Dice Score
Original Model (15 Epochs)	0.8074
Extended Training Model	0.8010

Figure 4.4 presents qualitative segmentation results obtained from the extended training model. The figure compares original histopathology images, ground truth masks with nuclei counts, and predicted segmentation masks with corresponding predicted nuclei counts. The results demonstrate that the model successfully segments most nuclei regions while maintaining accurate nuclei counting performance across different tissue samples.

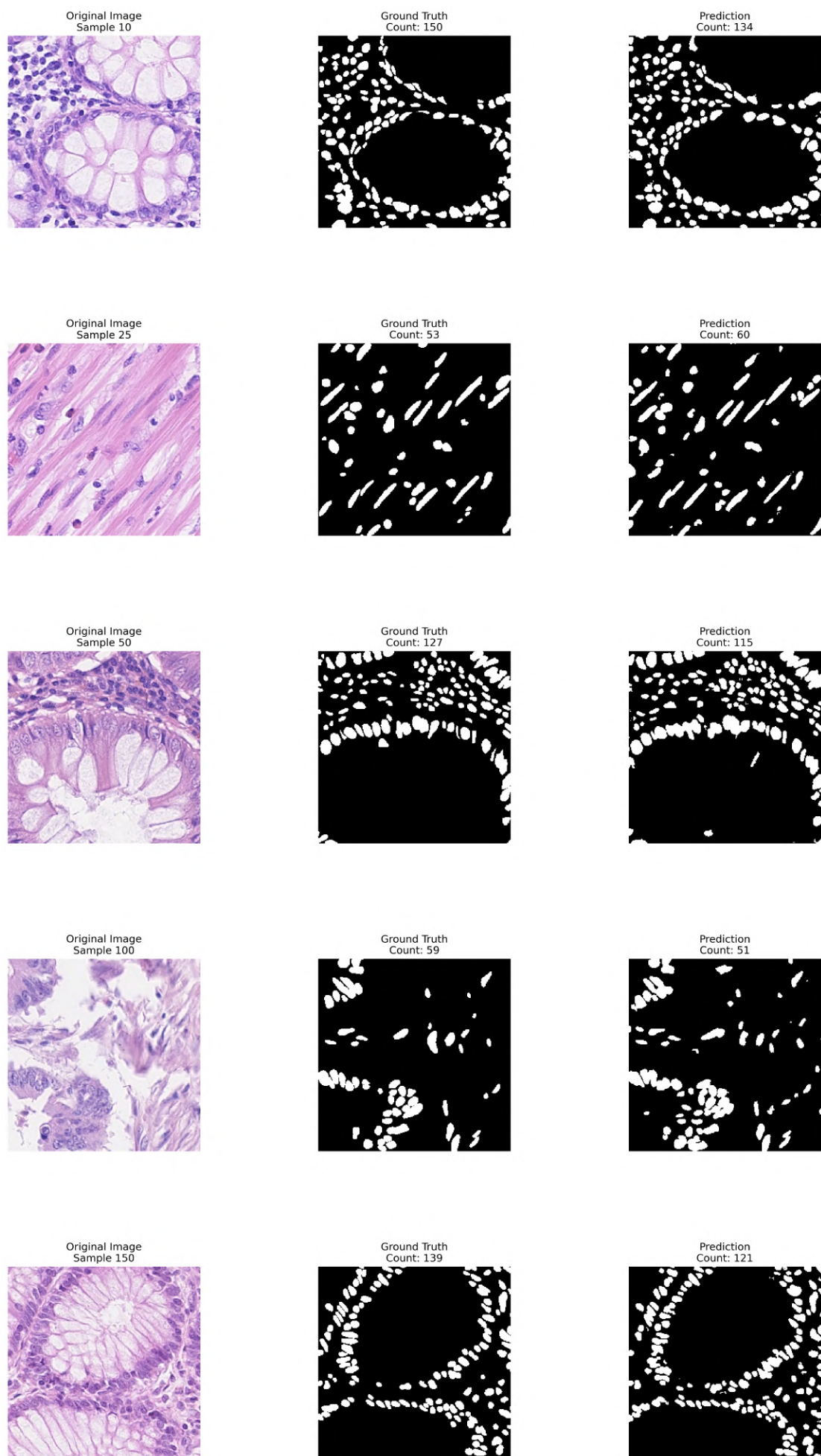


Figure 4.4: Segmentation and nuclei counting results after extended training

From the results, it can be observed that increasing training duration alone did not significantly improve segmentation performance. The slight reduction in Dice score indicates that the original model had already achieved near-optimal convergence.

This experiment demonstrates that architectural enhancements such as attention mechanisms, multi-scale feature extraction, and the JPFM bottleneck contributed more significantly to performance improvement than simply increasing training epochs.

4.7 Discussion

The experimental results show a clear progression from traditional segmentation methods toward advanced deep learning architectures.

Watershed segmentation performed poorly due to handcrafted limitations.

U-Net and U-Net++ significantly improved segmentation performance.

SANet demonstrated strong counting capability but lacked pixel-level precision.

Attention mechanisms improved feature localization.

The MultiScale Attention + JPFM enhancement provided the best overall segmentation performance by combining:

- Attention-based feature selection
- Multiscale feature extraction
- Dilated convolution based contextual learning

4.8 Summary

This chapter presented qualitative and quantitative analysis of all implemented models. The MultiScale Attention + JPFM model achieved the best overall performance and was selected as the final model for colon nuclei segmentation.

Chapter 5

Conclusion and Future Scope

5.1 Conclusion

This work focused on the problem of automated nuclei segmentation and counting in colon histopathology images using both traditional image processing techniques and deep learning approaches.

The study was conducted using the CoNIC dataset, which contains annotated histopathology images with pixel-level nuclei masks. Accurate nuclei segmentation plays an important role in computational pathology as it assists in disease diagnosis, cancer grading, and large-scale histological analysis.

Initially, a traditional watershed segmentation approach was implemented as a baseline method. Although it provided interpretable results, it suffered from severe over-segmentation and poor robustness under variations in staining and dense nuclei clusters.

To improve performance, multiple deep learning architectures were explored.

- U-Net was implemented as the first deep learning segmentation baseline.
- U-Net++ was explored to improve feature reuse through nested skip pathways.
- SANet was implemented for nuclei density estimation and counting.
- Attention U-Net++ was developed to improve localization of relevant nuclei regions.
- A final Multiscale Attention + JPFM inspired architecture was implemented using multiscale attention mechanisms and dilated convolutions.

Experimental evaluation demonstrated a clear improvement in segmentation performance across increasingly advanced architectures.

The final proposed architecture achieved the best segmentation performance with:

- $\text{IoU} = 0.7177$
- $\text{Dice Score} = 0.8074$
- $\text{Recall} = 0.8510$

These results demonstrate that incorporating multiscale contextual learning and attention mechanisms improves nuclei segmentation performance in complex histopathology images.

The work successfully establishes a comparative framework for analyzing multiple segmentation approaches for colon nuclei analysis.

5.2 Limitations

Despite achieving improved performance, the current work has several limitations.

- The models were trained for a limited number of epochs due to computational constraints.
- The dataset contains significant variability in staining patterns and nuclei density, which remains challenging.
- The final model improves segmentation quality but does not achieve the best Count MAE.
- The study focuses only on nuclei segmentation and counting and does not perform nuclei classification.
- Real-time deployment in clinical workflows was not explored.

5.3 Future Scope

Several extensions can be explored in future work.

- Integration of transformer-based segmentation architectures.

- Multi-class nuclei classification along with segmentation.
- Deployment of the trained model as a pathology assistance tool.
- Use of larger histopathology datasets for improved generalization.
- Exploration of self-supervised learning for reduced annotation dependency.
- Integration with whole slide image analysis pipelines.

The work can be extended further toward building practical AI-assisted pathology systems for clinical applications.

Chapter 6

Bibliography

This chapter lists the references, datasets, and research resources used during the project.

6.1 Dataset Reference

1. CoNIC Dataset:
<https://conic-challenge.grand-challenge.org/>

6.2 Research Papers

1. U-Net Paper:
<https://arxiv.org/abs/1505.04597>
2. U-Net++ Paper:
<https://arxiv.org/abs/1807.10165>
3. SANet Reference Paper:
<https://arxiv.org/abs/2009.12666>
4. Attention U-Net Paper:
<https://arxiv.org/abs/1804.03999>
5. Multi-module U-Net++ for Colon Cancer Histopathological Image Segmentation:
<https://www.nature.com/articles/s41598-025-13636-6>

6. Enhancing Nuclei Segmentation in Histology Images by Effectively Merging Task-Specific and Foundation Models:
<https://doi.org/10.33851/JMIS.2024.11.4.219>

6.3 Software Libraries

1. TensorFlow Documentation:
<https://www.tensorflow.org/>
2. OpenCV Documentation:
<https://opencv.org/>
3. NumPy Documentation:
<https://numpy.org/>
4. Matplotlib Documentation:
<https://matplotlib.org/>
5. Scikit-image Documentation:
<https://scikit-image.org/>